DSC5001 Statistical Machine Learning

Hong Kong AQHI High-Risk Hour Early Warning:

A Comparative Study of Classification Models on Hourly Monitoring Data

Team-XX

[Student Name 1] (UID) | [Student Name 2] (UID) | [Student Name 3] (UID) | [Student Name 4] (UID)

October 2025

Research Question & Motivation

Core Question

Can we reliably predict whether next-hour AQHI will reach high-risk level (>=7) using past observations?

How do different classifiers trade off between catching high-risk hours vs. controlling false alarms?

Three models compared: Logistic Regression, Random Forest, XGBoost

Why classification, not regression?

Public health advice is risk-level-based, not numeric -> classification matches real application

Motivation

EPD publishes AQHI hourly but forecasts only 3x/day (00:30, 10:30, 16:30)

Hourly-level early warning has room for improvement during rapid pollution spikes

AQHI time series has temporal dependency + severe class imbalance (<2% high-risk)

Scientific value: not chasing best accuracy, but analyzing model trade-offs

Missed alerts (FN) vs. false alarms (FP) have different real-world costs

Data Sources & Feature Engineering

Data Sources

AQHI Hourly Data (EPD / data.gov.hk)

18 stations: 15 general + 3 roadside (Causeway Bay, Central, Mong Kok)

Period: Jan 2021 - Dec 2024 (48 months, avoiding 2025 methodology change)

Total: 617,799 samples after preprocessing

Weather Data (HKO / data.gov.hk)

Daily mean temperature, relative humidity, rainfall, wet bulb temp

Merged by date with AQHI records

Feature Engineering (27 features)

Lagged AQHI: lag-1, 2, 3, 6, 12, 24 hours

Rolling stats: 3h/6h mean, 6h std

AQHI differences: 1-step and 3-step

Time encoding: hour, day-of-week, month (sin/cos)

Weather: temperature, humidity, rainfall, wet bulb

Station type: roadside vs. general (binary)

Target: next-hour AQHI >= 7 -> binary (1.60% positive)

Methodology

Three Classification Models

1. Logistic Regression (baseline)

L2 regularization, class_weight='balanced'

2. Random Forest

200 trees, max_depth=15, class_weight='balanced'

3. XGBoost (Gradient Boosting)

300 rounds, max_depth=6, scale_pos_weight

Evaluation Strategy

TimeSeriesSplit CV (5 folds)

Prevents future data leakage

Uses only past data to predict future

Class imbalance handling:

Threshold optimization via PR curve

No SMOTE (preserves temporal structure)

Key metrics:

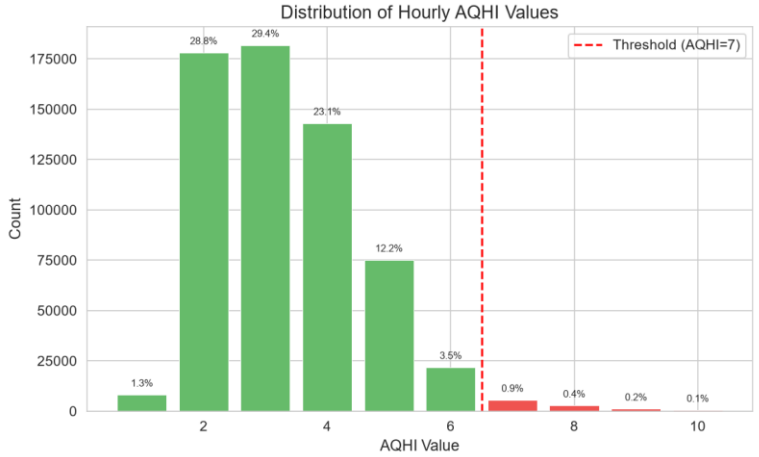
Recall (catch high-risk hours)

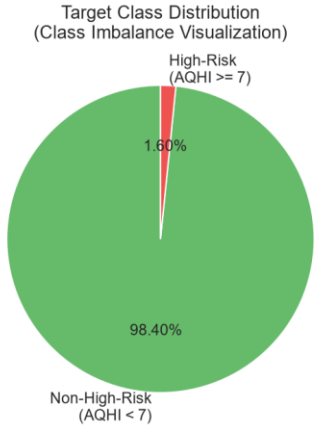
Precision (control false alarms)

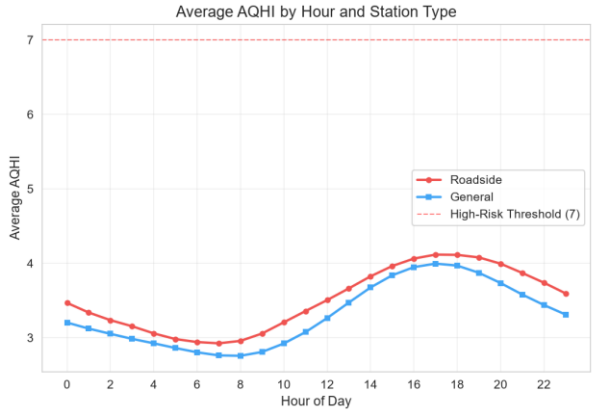
F1, PR-AUC, Confusion Matrix

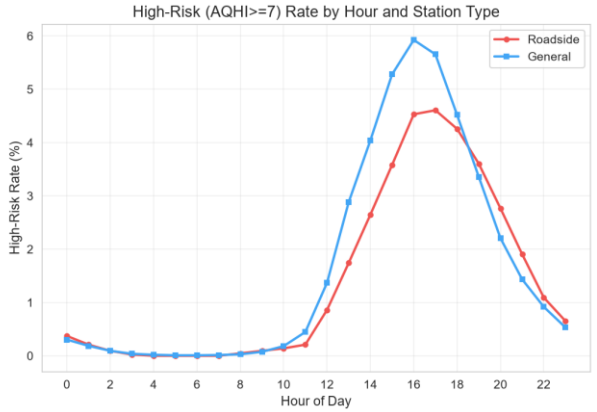
Cost analysis: FN cost = 5x FP cost

Data Exploration: AQHI Distribution & Patterns



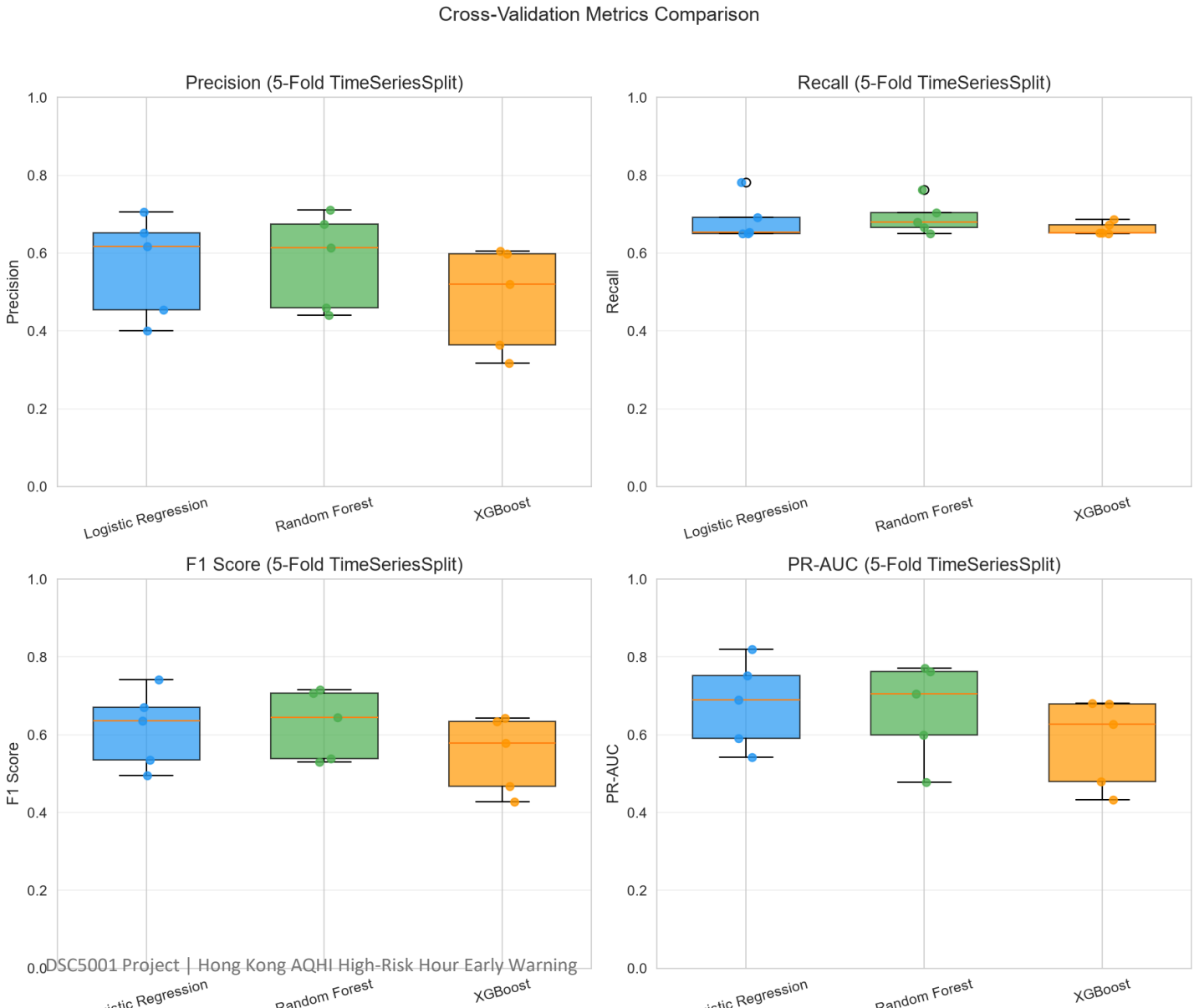






Key findings: AQHI is right-skewed with only 1.60% high-risk (>=7). Roadside stations show consistently higher AQHI than general stations. High-risk hours peak during afternoon (12:00-19:00) across both station types.

Cross-Validation Results (5-Fold TimeSeriesSplit)

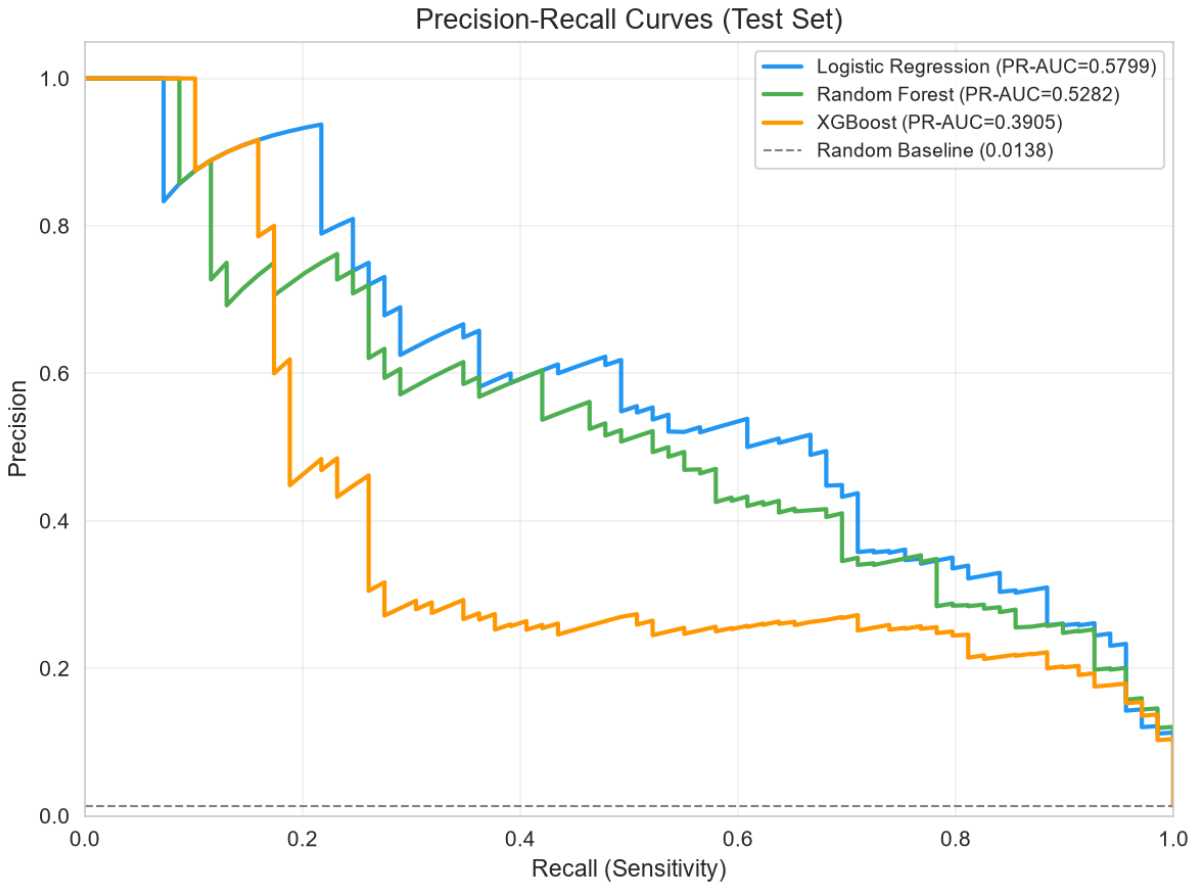


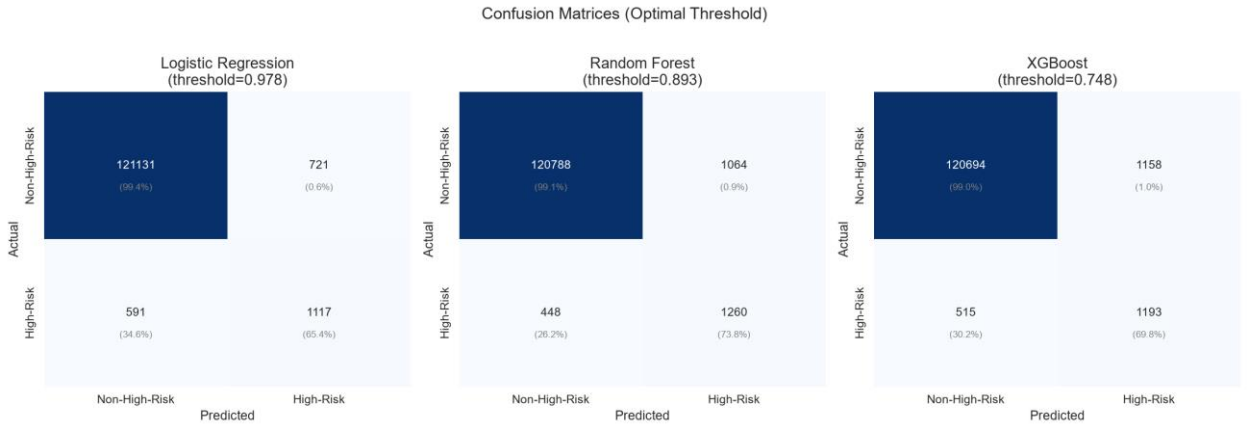
CV Summary (mean +/- std)

Model	Prec.	Recall	F1	PR-AUC
Log. Reg.	0.566 +/-0.118	0.686 +/-0.051	0.616 +/-0.090	0.679 +/-0.101
Rand. Forest	0.580 +/-0.111	0.693 +/-0.039	0.627 +/-0.080	0.663 +/-0.111
XGBoost	0.481 +/-0.119	0.663 +/-0.015	0.550 +/-0.088	0.580 +/-0.104

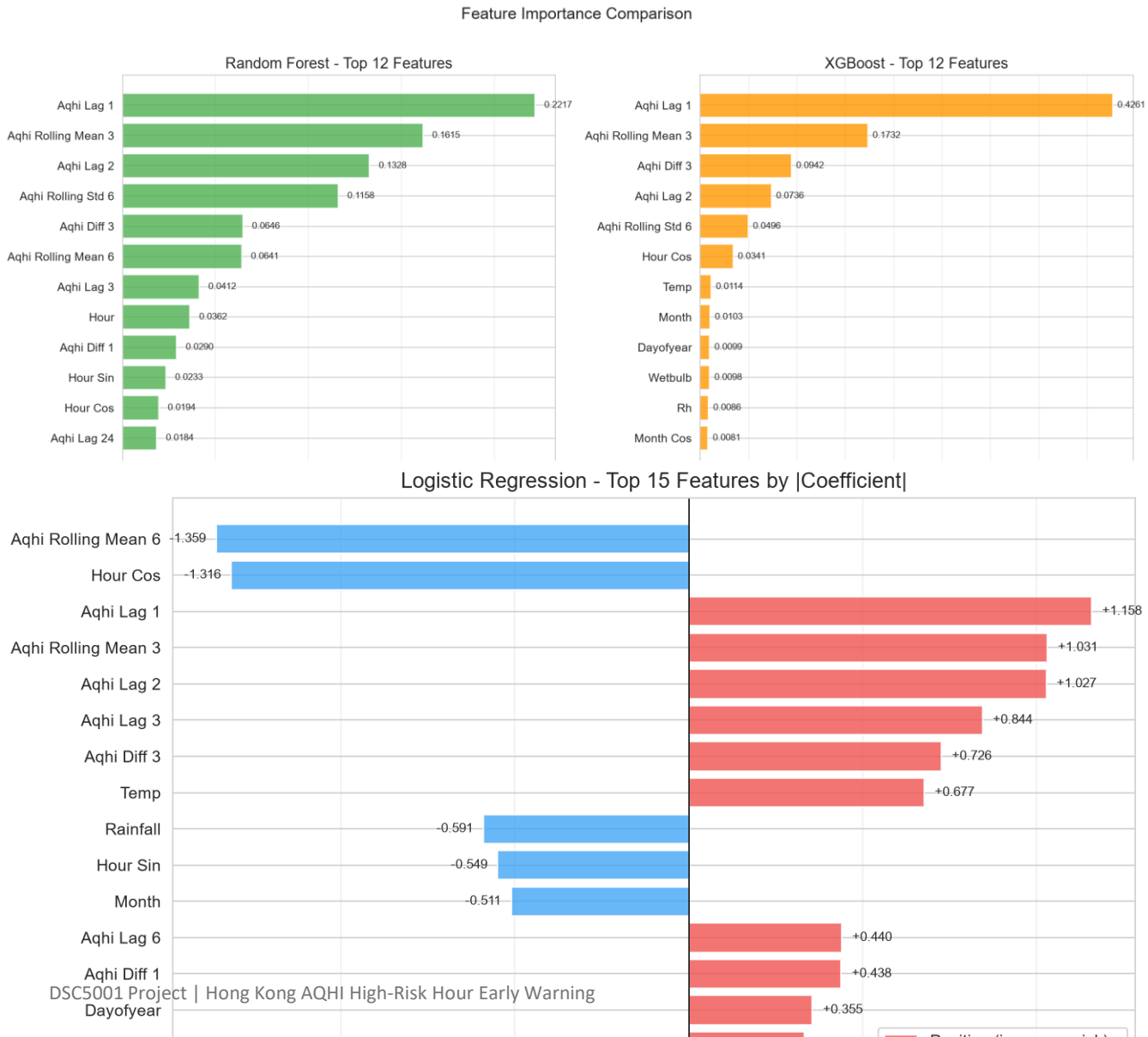
Logistic Regression and Random Forest perform comparably. XGBoost shows lower precision but more stable recall across folds.

Test Set Results: PR Curves & Confusion Matrices





Feature Importance Analysis



Key Findings

AQHI_lag_1 dominates all models

Most recent observation is strongest predictor

Rolling mean (3h) is 2nd most important

Recent trend captures momentum

Rolling std (6h) matters for RF

Volatility signals impending risk

Hour-of-day patterns are significant

Afternoon hours (12-19) show higher risk

Temperature has positive coefficient in LR

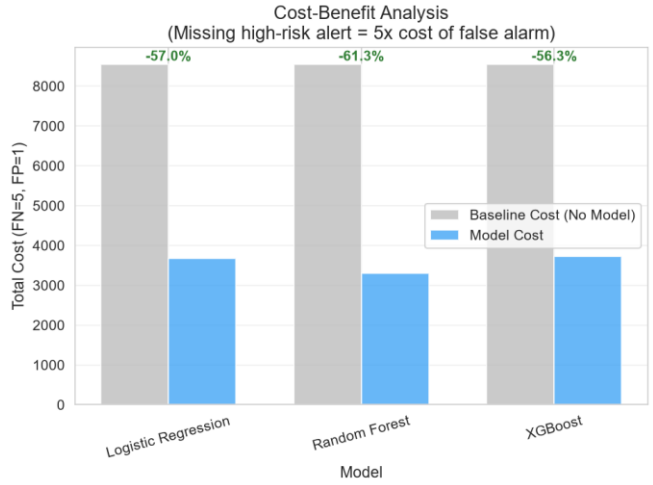
Higher temp -> more high-risk hours

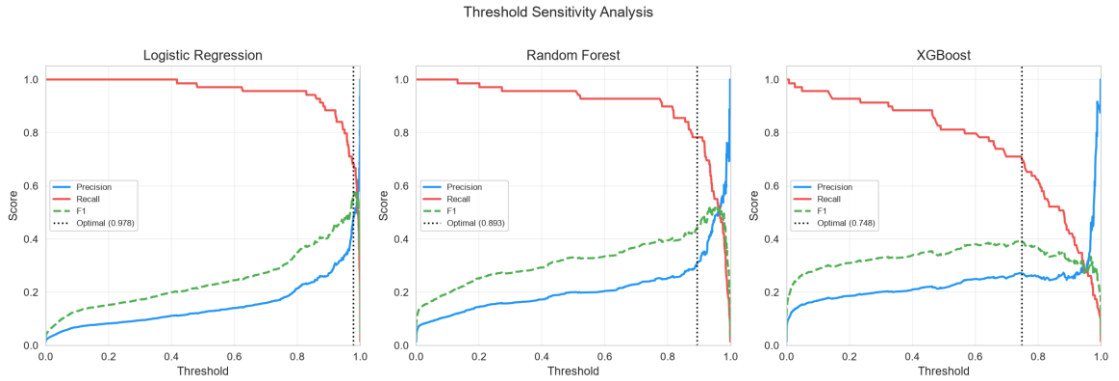
Rainfall has negative coefficient

Rain washes pollutants from air

Cost-Benefit Analysis & Threshold Optimization







Assumption: missing a high-risk alert (FN) costs 5x a false alarm (FP). Random Forest achieves the highest cost reduction (61.3%), with the fewest missed alerts (448) despite more false alarms. Optimal thresholds are model-specific: LR=0.978, RF=0.893, XGB=0.748.

Conclusions & Limitations

Conclusions

Classification framework is viable for AQHI early warning

Binary prediction matches public health advice structure

Random Forest: best recall (73.8%) & cost reduction (61.3%)

Recommended when missing high-risk hours is most costly

Logistic Regression: best precision (60.8%) & highest PR-AUC

Recommended when false alarms must be minimized

XGBoost: balanced but lower overall performance

May benefit from more hyperparameter tuning

Key predictor: AQHI_lag_1 (most recent observation)

Simple lagged features outperform meteorological variables

Limitations & Future Work

Weather data is daily, not hourly

Limits temporal resolution of meteorological features

Station spatial mismatch

EPD stations and HKO weather stations don't co-locate

AQHI methodology changed in March 2025

Used 2021-2024 data to avoid distribution shift

Class imbalance is severe (1.60%)

Even with weighting, precision remains moderate

Future: hourly weather data, multi-station spatial models

Deep learning (LSTM/Transformer) for sequence prediction